

## **Identifying Companies with Durable Competitive Advantage at an Early Stage**

With the advance of AI, this project explores whether it is possible to identify companies with durable competitive advantages at an early stage, before they are widely recognized by the public. Today, financial markets are highly competitive, and almost everyone has access to computers or smartphones, allowing stocks to be bought or sold within seconds. Stock prices often adjust immediately when new information is released, and in many cases, stocks are already trading at inflated prices before the information becomes widely available.

### **Main Goals**

The main goal of my project is to build a model that can identify companies with durable competitive advantages at an early stage. Before the public becomes aware of them, these companies' stocks would still trade at a significant discount. Many hedge funds already use similar strategies, but because these tools are widely accessible, it is possible to compete using a well-designed approach.

### **Initial approach**

At this point, it is difficult to know exactly how I will approach the project, but I have a basic plan. I will call various available APIs to gather financial and alternative data. For example, I could use the Securities and Exchange Commission (SEC) API to collect financial statements and other relevant filings. Machine learning can then be used to identify early patterns shared by historically successful companies, such as Amazon, Google, Microsoft, and Nvidia. Features could include revenue growth, patent portfolios, customer base, market share, and other key financial metrics. I could also analyze web-based sources to see what others are saying, including important news articles, and process earnings call transcripts to detect the tone of management, which may provide additional insights into the company's future prospects.